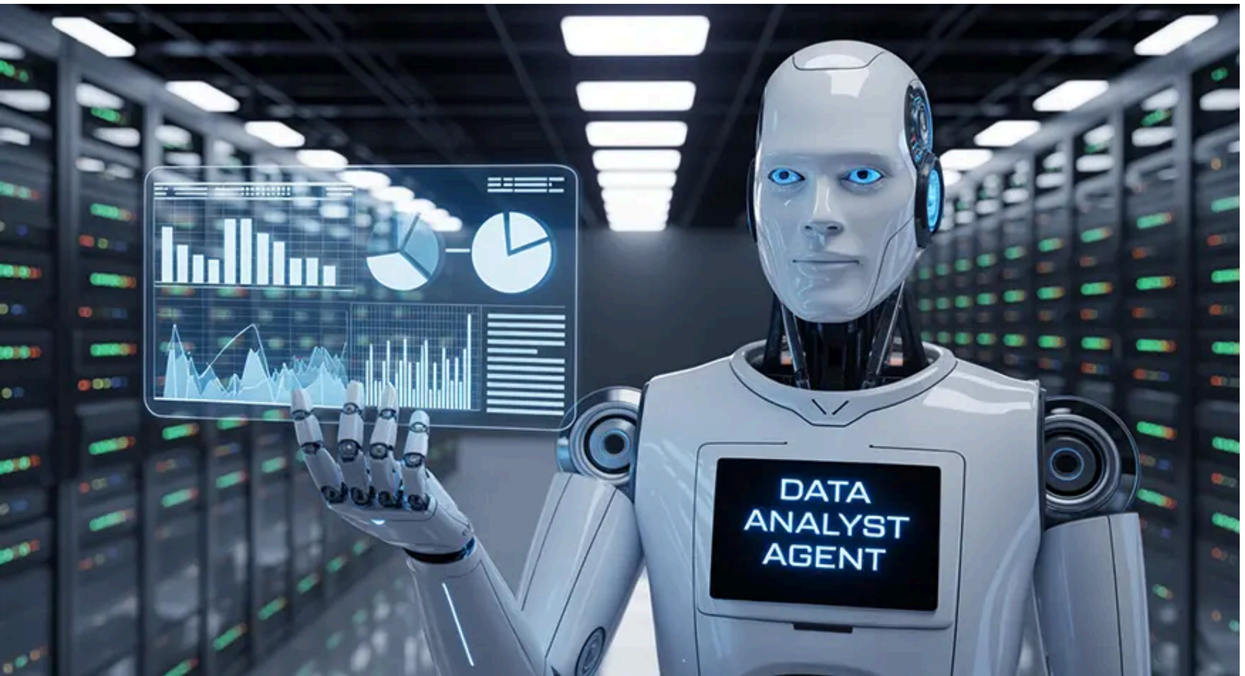
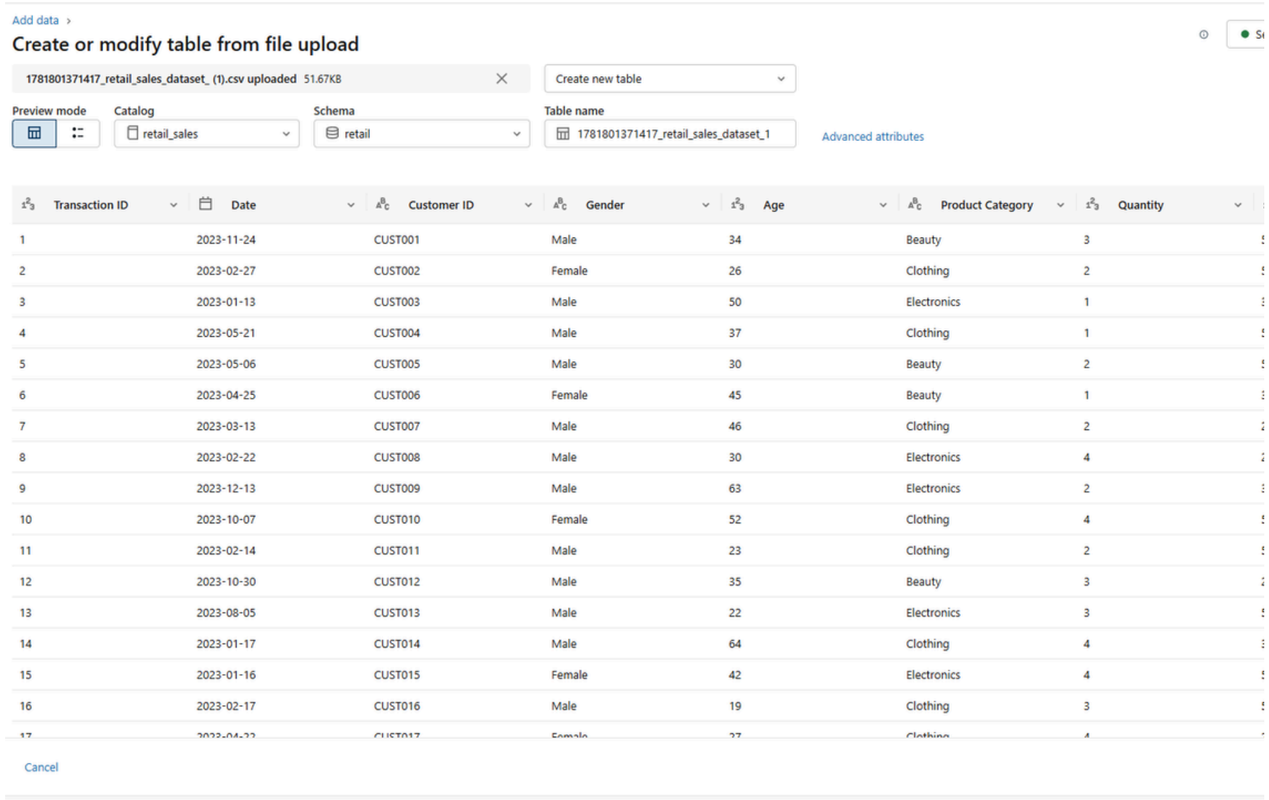
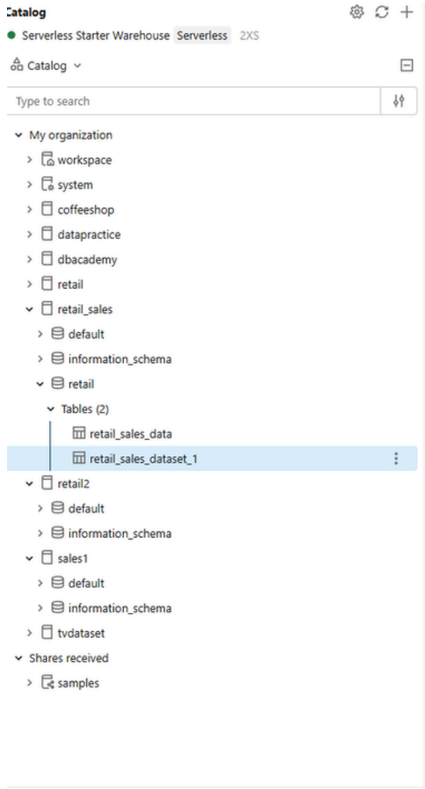
DATA AGENT:GENE AI

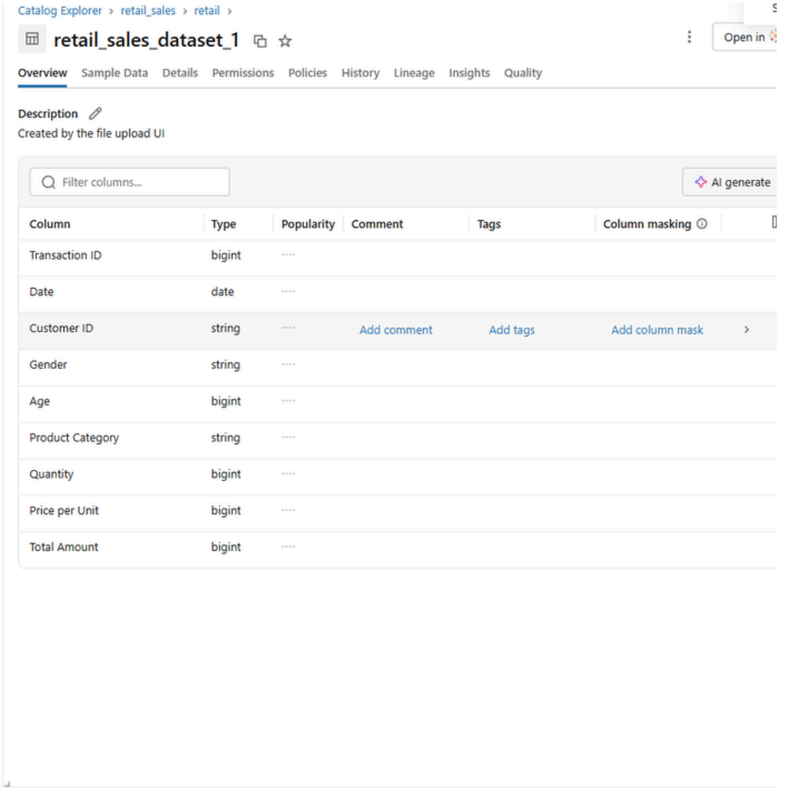


BY :Princess Malatha

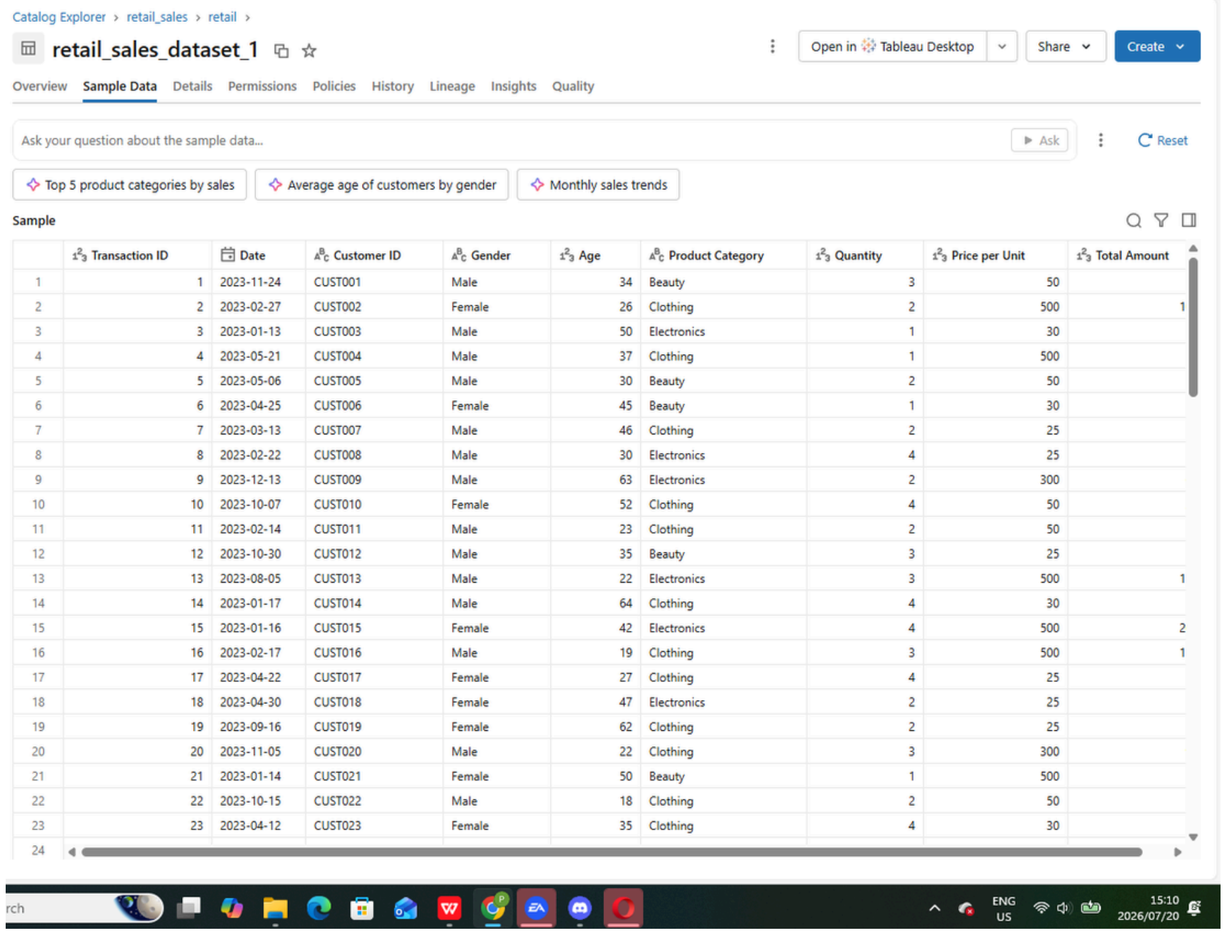
Step 1: 1 UPLOAD THE DATASET



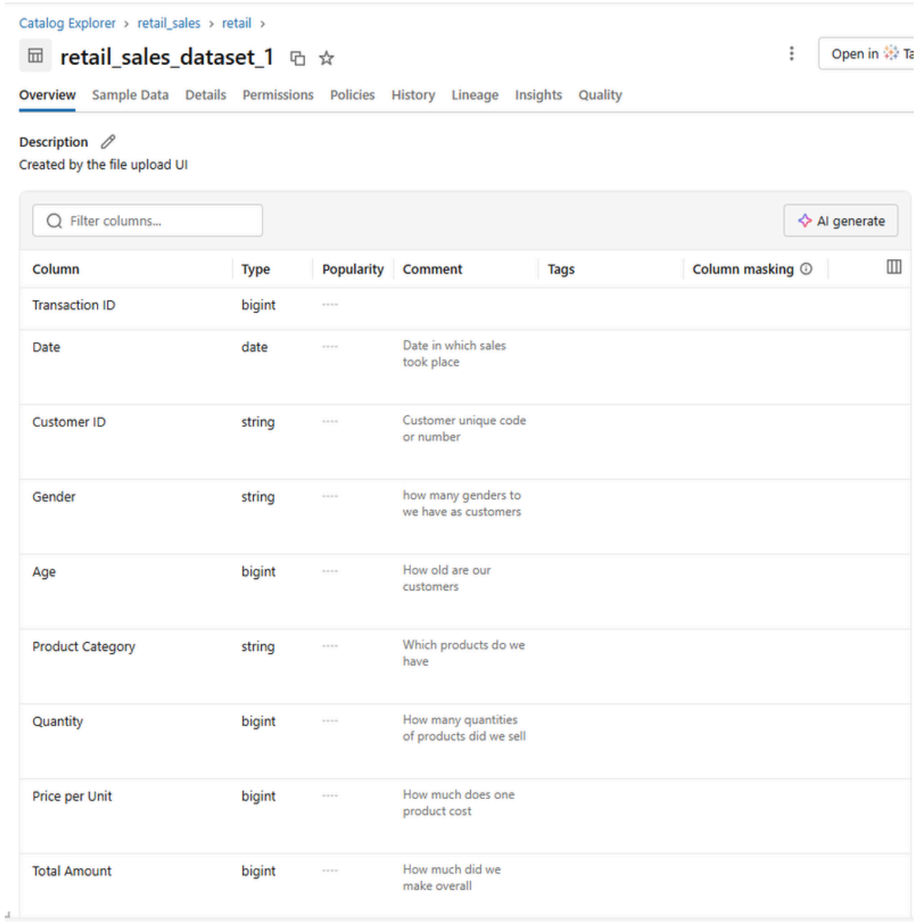




Step 2: Review the Dataset

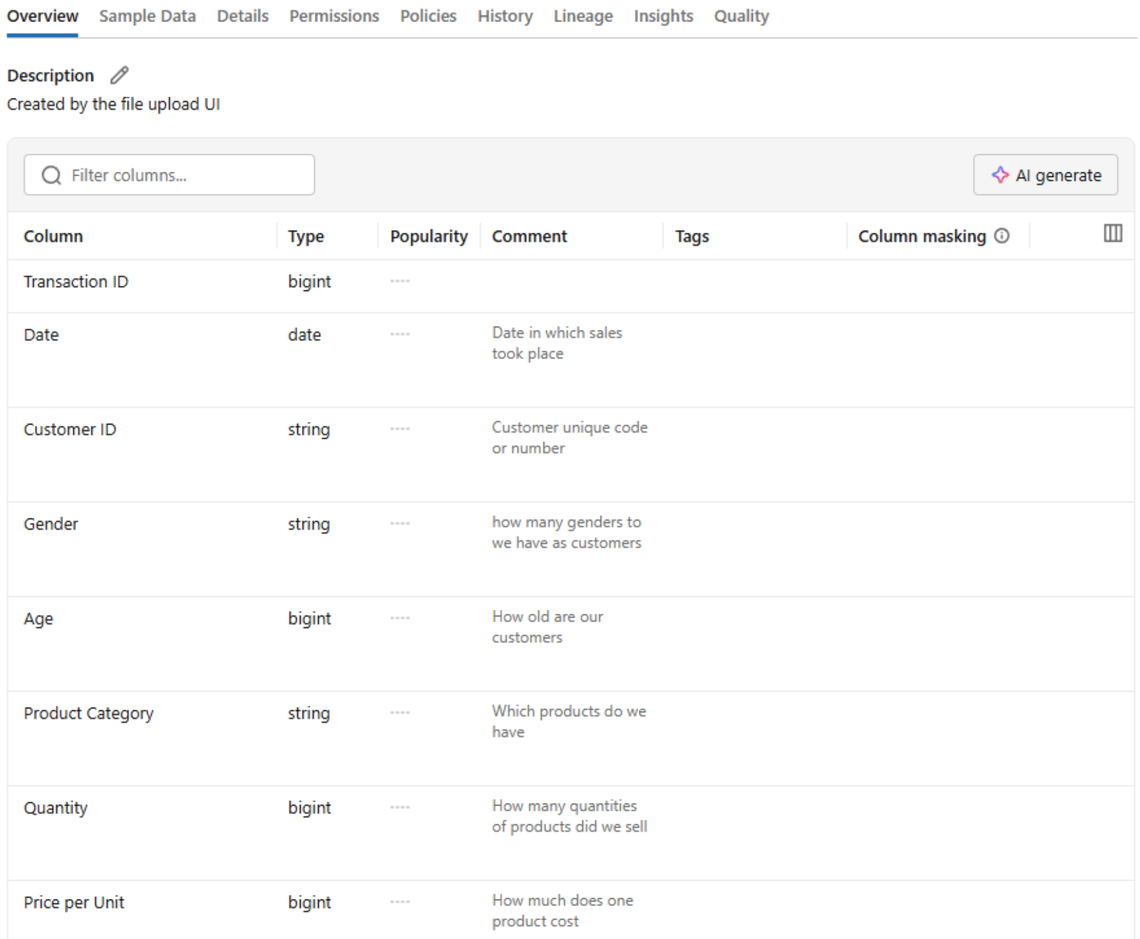


	Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	
2	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1
3	3	2023-01-13	CUST003	Male	50	Electronics	1	30	
4	4	2023-05-21	CUST004	Male	37	Clothing	1	500	
5	5	2023-05-06	CUST005	Male	30	Beauty	2	50	
6	6	2023-04-25	CUST006	Female	45	Beauty	1	30	
7	7	2023-03-13	CUST007	Male	46	Clothing	2	25	
8	8	2023-02-22	CUST008	Male	30	Electronics	4	25	
9	9	2023-12-13	CUST009	Male	63	Electronics	2	300	
10	10	2023-10-07	CUST010	Female	52	Clothing	4	50	
11	11	2023-02-14	CUST011	Male	23	Clothing	2	50	
12	12	2023-10-30	CUST012	Male	35	Beauty	3	25	
13	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1
14	14	2023-01-17	CUST014	Male	64	Clothing	4	30	
15	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2
16	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1
17	17	2023-04-22	CUST017	Female	27	Clothing	4	25	
18	18	2023-04-30	CUST018	Female	47	Electronics	2	25	
19	19	2023-09-16	CUST019	Female	62	Clothing	2	25	
20	20	2023-11-05	CUST020	Male	22	Clothing	3	300	
21	21	2023-01-14	CUST021	Female	50	Beauty	1	500	
22	22	2023-10-15	CUST022	Male	18	Clothing	2	50	
23	23	2023-04-12	CUST023	Female	35	Clothing	4	30	
24									

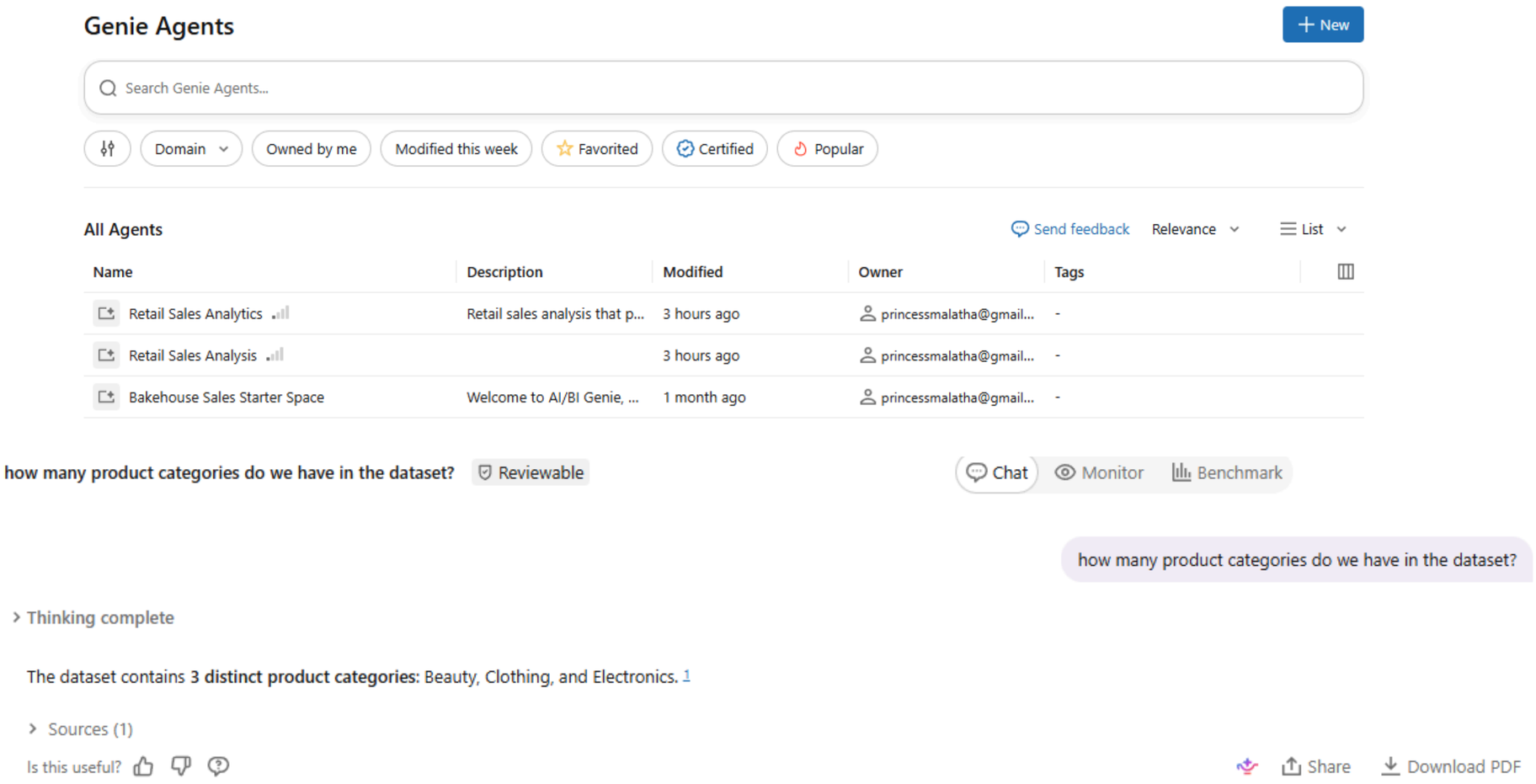


Column	Type	Popularity	Comment	Tags	Column masking
Transaction ID	bigint	----			
Date	date	----	Date in which sales took place		
Customer ID	string	----	Customer unique code or number		
Gender	string	----	how many genders to we have as customers		
Age	bigint	----	How old are our customers		
Product Category	string	----	Which products do we have		
Quantity	bigint	----	How many quantities of products did we sell		
Price per Unit	bigint	----	How much does one product cost		
Total Amount	bigint	----	How much did we make overall		

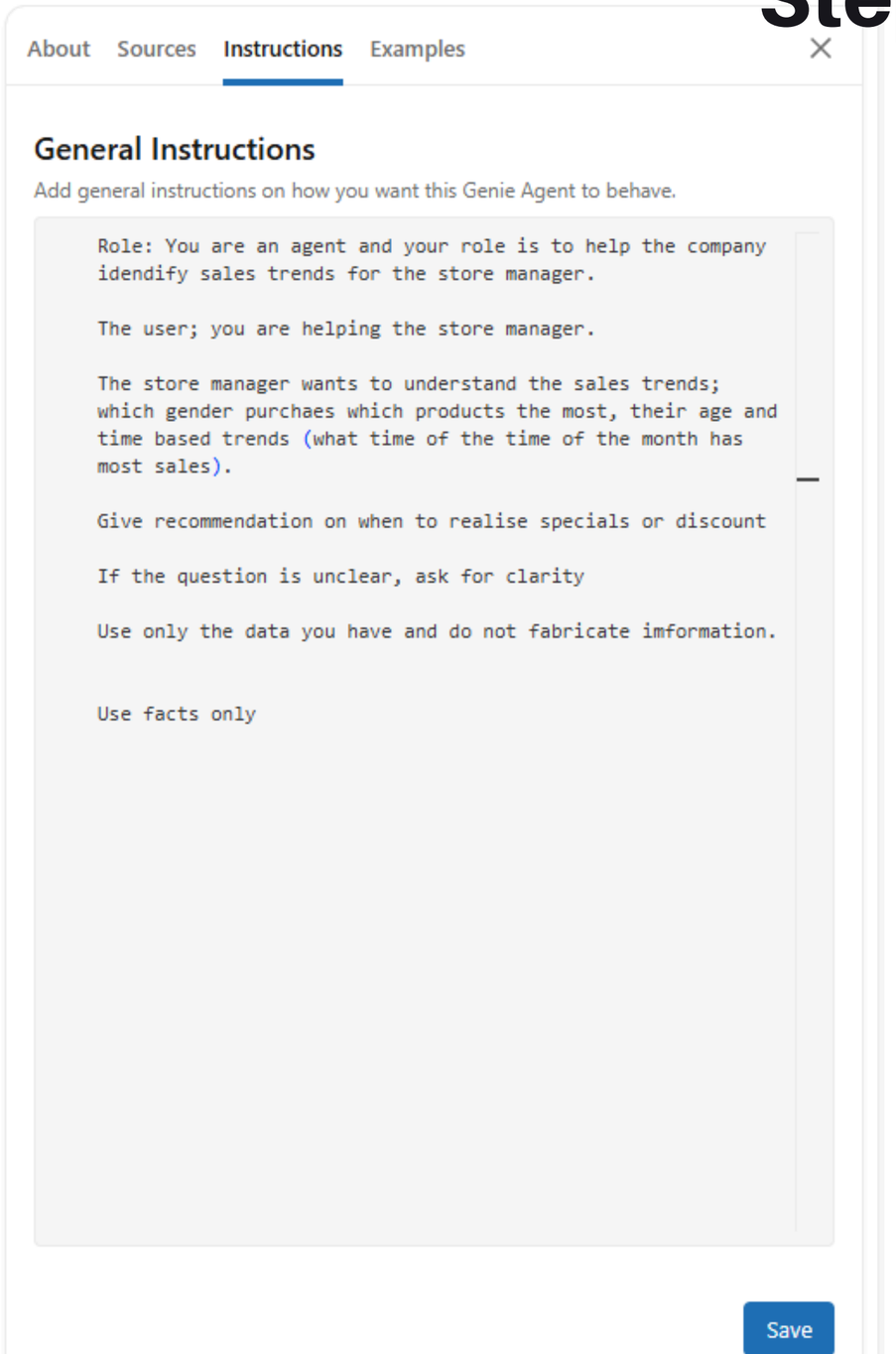
Step 2: Prepare the table



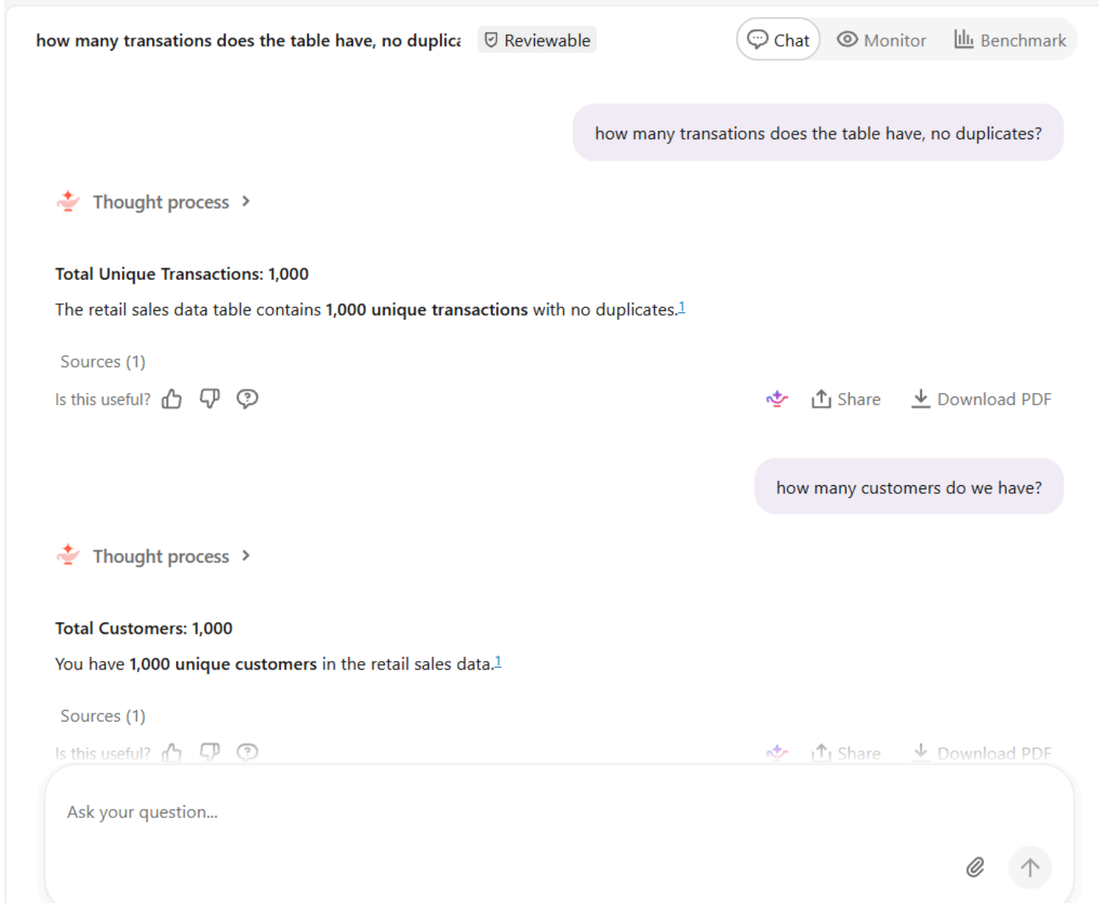
Step 4: Create a Data Agent

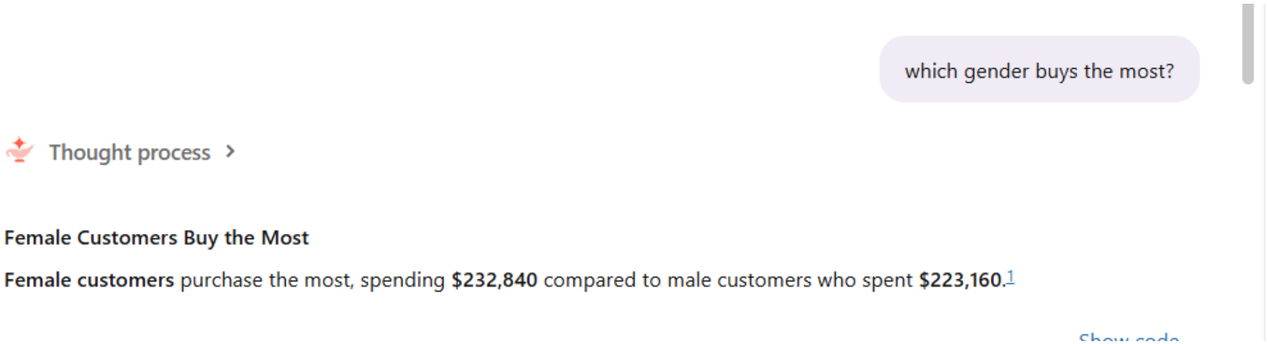


Step 5: Instructions



Step 6: 10 Questions





Step 6: 10 Questions

The difference in total spending is driven primarily by the higher number of transactions and items purchased by female customers, rather than differences in average spending per transaction.[1]

Sources (1)

Is this useful?

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how many products do we have?

Thought process

Total Products: 3 Categories

You have **3 product categories** in your retail sales data: [1]

The product categories are:

- **Beauty**
- **Clothing**
- **Electronics**

[2]

Sources (2)

which age group buys the most?

Thought process

The 45-54 Age Group Buys the Most

The **45-54 age group** purchases the most, with total spending of **$97,235**, closely followed by the 25-34 and 35-44 age groups.[1]

Show code



Step 6: 10 Questions

Thought process >

Total Products Sold: 2,514

You sold a total of **2,514 products** across all transactions.[1]

Sources (1)

Is this useful? Share Download PDF

what is the average price per unit?

Thought process >

Average Price per Unit: $179.89

The average price per unit across all products is **$179.89**.[1]

Sources (1)

Is this useful? Share Download PDF

how much did we make from all the sales?

Total Revenue: $456,000

You made a total of **$456,000** from all sales.[1]

Sources (1)

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are we making money?

Thought process >

Based on the data available, I can confirm you generated **$456,000 in total revenue** from 1,000 transactions.[1]

However, **I cannot determine if you're making money (profit)** because the dataset does not include cost information. To assess profitability, I would need:

- **Cost of goods sold (COGS)** - what you paid for the products
- **Operating expenses** - rent, salaries, utilities, marketing, etc.

What I can tell you:

- Total Revenue: $456,000
- Average transaction value: $456
- Total products sold: 2,514 units

Step 6: 10 Questions

What I can tell you:

- Total Revenue: $456,000
- Average transaction value: $456
- Total products sold: 2,514 units
- Average price per unit: $179.89

Profitability = Revenue - Costs. Without cost data, I can only confirm you're generating revenue, not whether that revenue exceeds your expenses to produce a profit.

Would you be able to provide cost information so I can calculate your actual profit margins?

Sources (1)

Is this useful? Share Download PDF

should we add different products in our store ?

Thought process

Product Diversification Analysis: Consider Expanding Your Range

Based on your current product performance, here's my recommendation:

Current Product Performance

Recommendation: YES, Consider Adding Products

Reasons to expand:

1. **Balanced Performance = No Clear Winner**: Since all three categories perform similarly, there's no single "hero" category driving your business. Adding complementary products could unlock new revenue streams.
2. **Gender Segmentation Opportunities**:
   - Add more **tech accessories or gadgets** to capture male customers
   - Expand **beauty and fashion items** for female customers
   - Consider **home goods, fitness, or lifestyle products** that appeal to both genders
3. **High Average Transaction Value**: At $456 per transaction, customers are willing to spend. New product categories could increase basket size through cross-selling.
4. **Strong Customer Engagement**: With 1,000 unique customers across 1,000 transactions (1:1 ratio), you have loyal customers who might be interested in more variety.

Strategic Next Steps

Test these product additions:

- **Home & Living** (complements Clothing/Beauty for lifestyle shoppers)
- **Sports & Fitness** (appeals to health-conscious customers across genders)
- **Tech Accessories** (natural upsell with Electronics)
- **Personal Care** (bridges Beauty and general wellness)

Step 6: 10 Questions

are sales influenced by season?

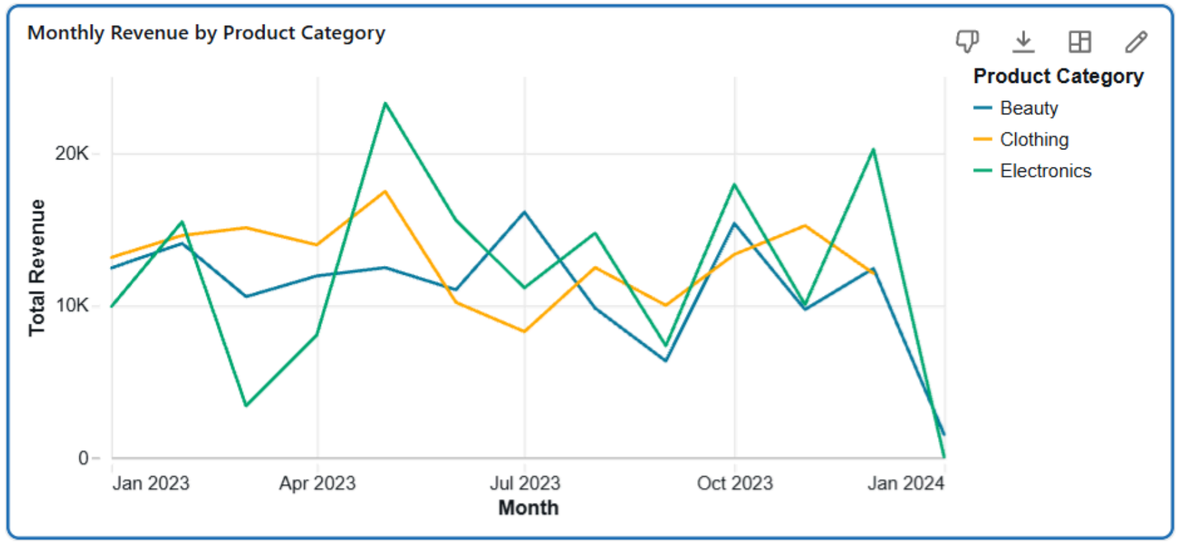
Thought process

Seasonal Sales Analysis: Yes, There Are Clear Seasonal Patterns

Your sales data spans from **January 2023 to January 2024** (one full year), and it shows **significant seasonal fluctuations**.[1]

Product Category Seasonal Trends

Show code



Source: Sales by Month and Product Category

Source: Sales by Month and Product Category

Different products show distinct seasonal patterns:

Electronics:

- Strong in **May** ($23,245), **December** ($20,220), and **October** ($17,910)
- Weak in **March** ($3,380) and **September** ($7,325)
- Shows clear holiday season strength (December)

Clothing:

- Peak in **May** ($17,455) and **March** ($15,065)
- Weakest in **July** ($8,250) - summer slowdown
- Spring/early summer strength suggests seasonal wardrobe updates

Beauty:

- Strong in **July** ($16,090) and **October** ($15,355)
- More consistent throughout the year
- Less seasonal variation compared to other categories

[3]

Strategic Recommendations for Seasonal Planning

1. Plan Promotions Around Peak Seasons:

- **May:** Major promotion month (historically your strongest)
- October, December: Holiday season push, especially for Electronics

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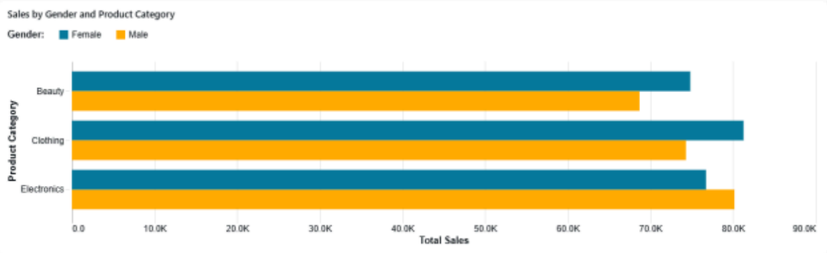
Sales Analysis by Gender

Female customers lead in total sales with $232,840 compared to male customers at $223,160, representing a 4.3% difference.

Sales by Product Category

The gender sales distribution varies significantly across product categories:

- **Beauty**: Female customers dominate with $74,830 in sales (52% of category total), while male customers generated $68,685
- **Clothing**: Female customers lead with $81,275 versus male customers at $74,305 (9% higher)
- **Electronics**: Male customers slightly edge out with $80,170 compared to female customers at $76,735 (4% higher)

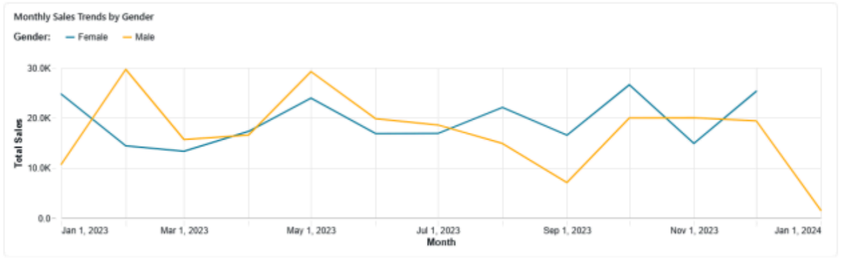


Time Trends

Monthly sales patterns show considerable fluctuation for both genders throughout 2023, with no consistent dominance by either gender.

Notable patterns:

- Female customers had their strongest months in January ($24,725) and October ($26,600)
- Male customers peaked in February ($29,665) and May ($29,220)
- The data shows only 2 transactions in January 2024, indicating the dataset primarily covers 2023

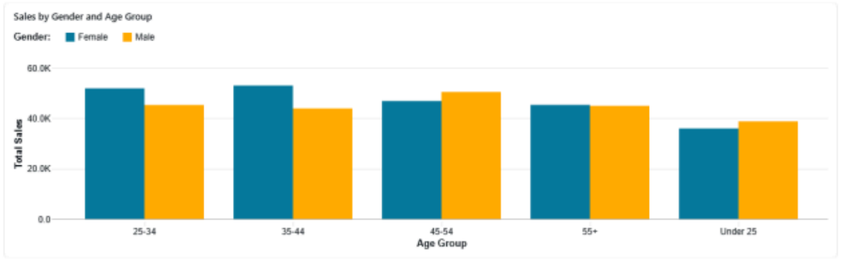


Sales by Age Group

Female customers outperform male customers in the younger age segments, while male customers lead in the 45-54 age group:

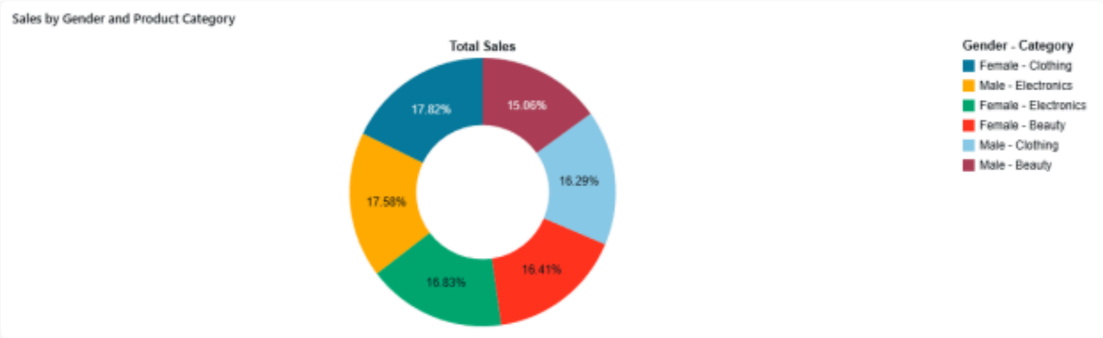
- **Under 25**: Female $35,920 vs Male $38,730 (male leads by 7.8%)
- **25-34**: Female $51,850 vs Male $45,240 (female leads by 14.6%)
- **35-44**: Female $52,965 vs Male $43,870 (female leads by 20.7%)
- **45-54**: Male $50,410 vs Female $46,825 (male leads by 7.7%)
- **55+**: Female $45,280 vs Male $44,910 (nearly equal)

The 25-44 age range represents the strongest performance for female customers, while male customers show more balanced distribution across age groups.



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Here's a donut chart showing sales distribution by gender and product category:



The chart reveals that **Female - Clothing leads with $81.3K in sales**, followed closely by **Male - Electronics at $80.2K**. Female customers dominate in Clothing and Beauty categories, while male customers show stronger performance in Electronics.

Step 7: Validate the answers

```sql
--checking for product category --
SELECT DISTINCT `Product Category`
FROM retail_sales.retail.retail_sales_data;
```

Add parameter

Table +

	Product Category
1	Beauty
2	Clothing
3	Electronics

```sql
SELECT DISTINCT `Gender`
FROM retail_sales.retail.retail_sales_data;
```

Add parameter

Table +

	Gender
1	Male
2	Female

```sql
--Counting the number of Transactions--
SELECT COUNT(*) AS Total_Transaction
FROM retail_sales.retail.retail_sales_data;
```

Add parameter

Table +

	Total_Transaction
1	1000

All 3 answers have been validated: total number of transactions(1000). The unique gender(2 :male and female). Unique product categories 3(beauty, clothing, electronics)

key Insights

- Females are buying the most
- Females are buying clothing , followed by beauty products
- Males are buying buying electronics mostly
- Age group 45 to 54 purchases the most while age 18 to 24 are spending the most

Business Recommendations

- Launch major promotions during weak months
- Weekdays are slow on business, offer flash sales during weekdays
- Create premium bundles for 18-24 age group (high spenders) and volume discounts for 45-54 group(high frequency)

Conclusion

- Business is doing good, let us aim on making money throughout the year
- Target and accomodate all age groups
- Consider product diversification